

Automatic Fan System

Arduino Uno + Tinkercad Simulation Project

Project Overview

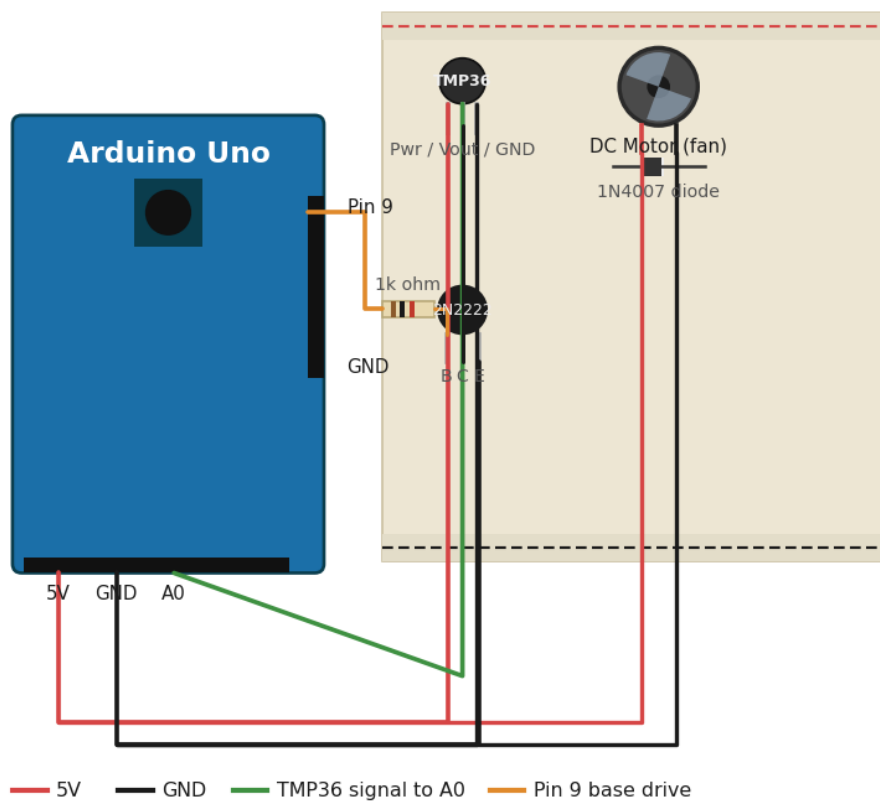
Yeh project TMP36 temperature sensor ka istemal karke ambient temperature ko continuously measure karta hai. Jab temperature ek fixed threshold (28 C) se upar chala jata hai, ek DC motor (fan) automatically ON ho jata hai taake cooling ho sake. Roshni thandi hone par fan wapas OFF ho jata hai. Motor ko directly Arduino pin se nahi chalaya jata - ek NPN transistor (2N2222) ise switch ki tarah control karta hai, aur ek flyback diode motor ke reverse voltage spikes se transistor ko protect karta hai.

Components Used

#	Component	Quantity	Purpose
1	Arduino Uno R3	1	Main microcontroller board
2	Breadboard	1	To hold and connect components
3	Temperature Sensor (TMP36)	1	Senses ambient temperature
4	DC Motor (fan)	1	Cooling fan actuator
5	NPN Transistor (2N2222)	1	Switches motor ON/OFF (Arduino pin can't supply motor current directly)
6	Diode (1N4007)	1	Flyback protection against motor's reverse voltage spike
7	Resistor - 1k ohm	1	Base resistor - limits transistor base current
8	Jumper wires	as needed	Connections between components

Note: Motor ko Arduino pin se direct nahi chalaya ja sakta kyunki motor zyada current kheenchta hai jo Arduino pin (max ~40mA) handle nahi kar sakta. Transistor ek switch ki tarah kaam karta hai - Arduino sirf base ko control karta hai, transistor motor ka bada current path switch karta hai.

Circuit Diagram (Tinkercad Style)



TMP36 wired to A0 (green), Arduino Pin 9 drives the transistor base through a 1k resistor (orange), and the transistor switches the motor's ground path. Red = 5V, Black = GND.

Wiring Information

From	To	Wire Color (suggested)
TMP36 Power	Arduino 5V	Red
TMP36 GND	Arduino GND	Black
TMP36 Vout	Arduino A0	Green
Arduino Pin 9	1k resistor - leg 1	Orange
1k resistor - leg 2	Transistor Base (B)	Direct breadboard row
Transistor Emitter (E)	Arduino GND	Black
Transistor Collector (C)	Motor negative terminal	Direct breadboard row
Motor positive terminal	Arduino 5V	Red
Diode cathode (banded end)	Motor positive terminal / 5V	Direct breadboard row

From	To	Wire Color (suggested)
Diode anode	Motor negative terminal / Collector	Direct breadboard row

Diode direction is important: diode ka banded (cathode) end 5V/motor-positive ki taraf hona chahiye, aur anode collector/motor-negative ki taraf - reverse lagane se protection kaam nahi karegi.

Step-by-Step Build Procedure

Step 1: Naya circuit banao

Tinkercad kholo, "Circuits" section mein jao, aur "Create new Circuit" par click karo.

Step 2: Components add karo

Right-side panel se ye cheezein drag karke canvas par lao: Arduino Uno R3, Breadboard, Temperature Sensor (TMP36), DC Motor, 2N2222 Transistor, 1N4007 Diode, aur 1k Resistor.

Step 3: TMP36 sensor wiring karo

TMP36 ka Power pin Arduino ke 5V se, GND pin Arduino ke GND se, aur Vout pin Arduino ke A0 (analog) pin se wire karo.

Step 4: Transistor base wiring karo

Transistor ka Base pin, 1k resistor ke through Arduino Pin 9 se jodo.

Step 5: Transistor emitter aur collector wiring karo

Transistor ka Emitter Arduino GND se jodo. Transistor ka Collector, Motor ke negative terminal se jodo.

Step 6: Motor wiring karo

Motor ka positive terminal Arduino 5V se jodo.

Step 7: Flyback diode lagao

Diode ko motor ke dono terminals ke parallel mein lagao - banded (cathode) end 5V/positive side par hona chahiye.

Step 8: Code likho aur simulation chalao

Arduino ko select karo, "Code" button dabao, Text mode select karo, aur neeche diya gaya pura sketch paste karo. "Start Simulation" dabao aur TMP36 ka temperature slider adjust karke fan ka behavior dekho.

Complete Arduino Sketch

Yeh code neeche pura selectable text hai - directly copy karke Tinkercad ya Arduino IDE mein paste kiya ja sakta hai.

```
// Automatic Fan System using TMP36
// TMP36 -> A0 | Motor (via transistor) -> Pin 9

const int sensorPin = A0;
const int fanPin = 9;
const float thresholdTemp = 28.0; // Celsius - fan ON hone ka threshold

void setup() {
  pinMode(fanPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int rawValue = analogRead(sensorPin);
  float voltage = rawValue * (5.0 / 1023.0);
  float temperatureC = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" C");

  if (temperatureC > thresholdTemp) {
    digitalWrite(fanPin, HIGH); // Fan ON
    Serial.println("Fan: ON");
  } else {
    digitalWrite(fanPin, LOW); // Fan OFF
    Serial.println("Fan: OFF");
  }

  delay(1000);
}
```

Kaise Kaam Karta Hai

TMP36 ambient temperature ko analog voltage mein convert karta hai, jo Arduino A0 par read hota hai aur formula $(\text{voltage} - 0.5) \times 100$ se Celsius mein convert hota hai.

Jab temperature threshold (28 C) se upar jata hai, Arduino Pin 9 ko HIGH karta hai. Ye signal 1k resistor ke through transistor ke Base tak pohanchta hai, jisse transistor ON ho jata hai aur Collector-Emitter ke through motor ka circuit complete ho jata hai - fan chalne lagta hai. Jab temperature threshold se neeche aata hai, Pin 9 LOW ho jata hai, transistor OFF ho jata hai, aur fan ruk jata hai.

Diode motor ke coil mein stored energy ko safely discharge karta hai jab motor achanak OFF hota hai - iske bagair ye reverse voltage spike transistor ko damage kar sakta hai.

Note: Agar fan simulation mein turant ON/OFF flicker kare, threshold value ko thoda adjust karo. Agar fan bilkul na chale, transistor ke B/C/E pin orientation aur diode ki direction dobara check karo.

Tinkercad Circuit Screenshot

